

Register 1 Blueprint Extension: The Asymmetric Density Filter

Natasha Zink

Heptagonal Unitary Field Theory

August 23, 2026

Abstract

This document outlines the theoretical expansion of the Minkowski M^4 Spacetime Matrix Dynamics framework. By analyzing the stark contrasts in JWST spectral data—specifically the chaotic hydrogen emission spectrum (GLIMPSE-17775) versus the highly structured methane absorption bands (WD 1856 b)—we formalize the **Information Density Filtering Law**. This blueprint is designed to be integrated into the continuous FDTD simulation architecture.

1 The Syntax Gate and Topological Backfill

The macroscopic universe operates as a rigid, syntax-driven 12-point voxel swap engine governed by Lie algebra. We observe that varying informational densities handle topological stress differently:

- **High-Capacitance Shunts (Light Data):** Lower-density informational substrates, such as uncompactified M^4 gas fields (Hydrogen), function as dissipation shunts. These fields absorb the violent differential backfill of 7-dimensional toroidal mass-energy injections (Φ_{7D}), resulting in the extensive electron scattering observed in the GLIMPSE-17775 peak.
- **Discrete Chromatic Filters (Heavy Data):** Conversely, highly compactified molecular hardware structures (e.g., hydrocarbon chains like Methane) act as rigid syntax gates. Their structural density permits only resonant Planck-tick clock cycles to pass without degradation, maintaining sharp, stable spectral lines.

2 The Dynamic Coupling Tensor Formulation

To operationalize this asymmetry in our PyMeep FDTD simulations, the scalar boundary condition $\kappa(r)$ is expanded into a coupling decay tensor $\kappa_{\mu\nu}(r)$. This allows us to map the topological stress directly into the material susceptibility (χ) grid parameters of the computational cell:

$$\kappa_{\mu\nu}(r) = \kappa_{M^4}\delta_{\mu\nu} + \left(\frac{\Phi_{7D}}{r^2 + \epsilon^2}\right)\sigma_{\text{scatter}}T_{\mu\nu} \quad (1)$$

Where:

- κ_{M^4} is the baseline stable coupling constant derived from exoplanetary spectra.
- Φ_{7D} represents the localized energy injection from the T^7 pinch point.
- r is the radial distance from the topological singularity.
- $T_{\mu\nu}$ represents the Lie algebra stress-energy generator acting upon the voxel collection.

3 Next Session Protocol

The following steps are slated for implementation in our FDTD environment:

1. Map the $\kappa_{\mu\nu}(r)$ tensor directly into the PyMeep `Simulation` cell as a spatially dependent susceptibility function.
2. Calibrate the maximum adjacent swap rate (e) against the FWHM velocity dispersion isolated from the JWST data.
3. Execute the GitHub Actions `stress_test.yml` continuous integration workflow against active 7D pinch geometries to ensure matrix stability.